

SQL Data Analysis Assignment

Keffi Tech Sales - Final SQL Analytics & Sales Performance Report

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Target Organization: Keffi Tech Sales

Scope: Comprehensive Analytical Report covering Data Quality Audit, Performance Metrics, Ranking, and Evidence-Based Business Insights based on SQL Transactional Data Analysis.

Executive Summary: This analytical report evaluates the sales performance of Keffi Tech Sales using SQL querying techniques applied across transaction logs, customer records, and product category metadata. The objective of this project is to transform raw transactional logs into actionable strategic intelligence, resolving data quality challenges (such as missing values and implied attributes) while answering critical business questions for executive management.

Section 1: Database Schema & Data Quality Audit

1.1 Schema Overview & Data Types

The transactional database consists of 1,500 total rows structured under the following schema definition:

Column Name	Data Type	Description / Analytical Note
Sales_ID	nvarchar	Unique transaction identifier (e.g., S10001, S10002).
Sales_Date	datetime2	Timestamp of transaction completion.
Customer_Name	nvarchar	Full name of the purchasing customer.
Gender	nvarchar	Customer gender (Contains NULL values; mapped to 'Unknown').
Country	nvarchar	Geographic origin of the sale.

Category	nvarchar	Product category (Electronics, Furniture, Clothing, Groceries). Missing values inferred via Unit_Price.
Product	nvarchar	Specific item purchased. Missing values inferred via Unit_Price mapping.
Quantity	nvarchar	Number of units purchased (Target for imputation / null handling).
Unit_Price	int	Monetary value per unit item (in NGN).
Payment_method	nvarchar	Channel used for payment (Cash, Card, Bank Transfer, Mobile Money).

1.2 Data Cleaning & Missing Value Treatment (Requirements 18 & 19)

Before conducting analytical queries, a thorough data quality audit was conducted to address missing values and potential duplicates:

- a. Gender Imputation:** Missing entries in the Gender attribute were imputed with the string label 'Unknown' to maintain dataset completeness without skewing demographic proportions.
- b. Product & Category Inference:** Missing entries in Product and Category were resolved using the fixed Unit_Price mapping (e.g., 12000 attributed to T shirt in Clothing Category)
- c. Quantity Treatment:** Records containing NULL values in Quantity are left the way it were.
- d. Duplicate Investigation:** Duplicate checks were performed across combined transaction composite keys (Sales_Date, Customer_Name, Unit_Price, Payment_method). No duplicate Found.

Section 2: Sales Performance & Key Analytical Findings

2.1 Overall Performance Metrics

Across the 1,500 recorded transactions in the database, overall financial performance is summarized below:

- a. **Total Transactions:** 1,500 transactions logged.
- b. **Total Units Sold:** 8,294 units (based on average order quantity of 5.5 units per transaction).
- c. **Total Revenue Generated:** NGN 920,293,000
- d. **Average Revenue Generated:** NGN 615,169

2.2 Top 5 Customers by Total Spending

The top 5 high-value customers contribute a substantial percentage of total company revenue.

Spending Rank	Customer Name	Total Spending
1	Joy Emmanuel	N 64,230,500
2	Grace Musa	N 57,234,000
3	Ruth Adams	N 55,439,000
4	Michael James	N 53,287,500
5	Mary Johnson	N 51,632,500

2.3 Category Performance & Revenue Contribution

Products were grouped into four core categories. The Electronics category dominated overall revenue contribution due to high unit prices

Category	Total Revenue	Revenue Percentage
Electronics	529600000	57.55%
Furniture	230625000	25.06%
Clothing	85492000	9.29%
Groceries	74576000	8.10%

2.4 Preferred Payment Methods

Cash was the most frequently used payment method, with 389 transactions and ₦245.458 million in revenue. Mobile Money followed with 383 transactions and ₦240.074 million, while Card recorded 374 transactions and ₦214.537 million. Bank Transfer recorded 354 transactions and ₦220.225 million.

The relatively close distribution across the four payment methods indicates that customers use both traditional and digital payment channels. The company should therefore continue supporting multiple payment options rather than relying on a single channel.

Payment preferences display a balanced distribution across digital and traditional payment channels:

Payment Method	Number Of Transaction	Total Revenue
Cash	389	₦245,458,000
Mobile Money	383	₦240,074,000
Card	374	₦214,536,500
Bank Transfer	354	₦220,224,500

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Section 3: Detailed SQL Solutions to Business Requirements

3.1 Monthly Sales Trends & Growth Analysis

Monthly performance: Revenue peaked in March at 96.356 million, followed by a sharp 26.17% decline in April. Revenue then recovered through September, but declined again in October–December, with December recording the largest monthly decline of 29.78%. This suggests that the business experienced significant fluctuations in monthly revenue and should investigate the causes of the recurring year-end decline.

Sales Year	Sales Month	Total Revenue	Previous Month Revenue	Revenue Growth Percentage
2025	1	59686500	NULL	NULL
2025	2	73461000	59686500	23.080000000000
2025	3	96356000	73461000	31.170000000000
2025	4	71141500	96356000	-26.170000000000
2025	5	77155000	71141500	8.450000000000

2025	6	76370500	77155000	-1.020000000000
2025	7	81236000	76370500	6.370000000000
2025	8	81640000	81236000	0.500000000000
2025	9	89715000	81640000	9.890000000000
2025	10	88611000	89715000	-1.230000000000
2025	11	73386000	88611000	-17.180000000000
2025	12	51534500	73386000	-29.780000000000

3.2 Above-Average Products and Customers

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The analysis compared individual product revenue against the average revenue generated across all products. The results show that Laptop, Smartphone, and Tablet are the strongest revenue-generating products. Laptop generated the highest product revenue at ₦238.05 million, followed by Smartphone at ₦167.44 million and Tablet at ₦101.70 million.

This indicates that high-value electronic products are major contributors to overall business revenue. Management should therefore maintain adequate stock levels for these products and monitor their demand closely.

Customers Generating Above-Average Revenue

Customer spending analysis identified several customers whose total spending was above the overall customer average. Joy Emmanuel was the highest-value customer, spending ₦64.23 million, followed by Grace Musa (₦57.23 million), Ruth Adams (₦55.44 million), Michael James (₦53.29 million) and Mary Johnson (₦51.63 million).

These customers represent important high-value accounts and should be targeted with customer-retention strategies, personalized offers and loyalty incentives.

3.3 Customer Purchasing Frequency

Customer purchasing frequency was assessed using the number of transactions recorded for each customer. Joy Emmanuel recorded the highest purchasing frequency with 88 transactions, followed by Grace Musa with 86 transactions and Mary Johnson with 84 transactions.

The results indicate that frequent customers are not necessarily identical to the customers with the highest spending. This distinction is important because purchasing frequency can be used to identify loyal customers, while total spending identifies high-value customers.

3.4 Sales Performance by Country

The country analysis was used to compare sales activity across the geographical markets represented in the dataset, Nigeria generated the highest revenue. This analysis helps management identify stronger markets and determine where sales resources and marketing activities should be concentrated.

Country	Total Transactions	Units Sold	Total Revenue
Nigeria	247	1405	₦175,135,500
South Africa	247	1351	₦162,215,500
United States	254	1423	₦160,658,500
Kenya	266	1431	₦155,531,000
Ghana	246	1363	₦147,331,500
United Kingdom	240	1321	₦119,421,000

3.5 Gender-Based Purchasing Patterns

Gender analysis showed relatively balanced purchasing activity between male and female customers. Female customers accounted for 758 transactions and ₦467.142 million in revenue, while male customers accounted for 738 transactions and ₦451.860 million. There were only 4 transactions classified as Unknown, generating ₦1.291 million.

Female customers therefore generated slightly higher transaction volume and revenue than male customers. However, the difference is relatively small, suggesting that both customer groups are important to the business.

3.6 Highest-Value Transactions

The highest-value transactions were identified by calculating transaction revenue as quantity multiplied by unit price. The highest transactions were ₦4.5 million, generated from purchases of 10 laptops at ₦450,000 each. Several customers recorded transactions at this value, including John Williams, Michael James, Mary Johnson, Joy Emmanuel and others. This

demonstrates the significant revenue impact of high-value electronics transactions, particularly laptops.

Sales_ID	Customer_Name	Product	Quantity	Unit_Price	Revenue	Sales_Date
S10050	John Williams	Laptop	10	₦ 450,000	₦ 4,500,000	2025-02-20
S10205	Michael James	Laptop	10	₦ 450,000	₦ 4,500,000	2025-03-29
S10380	Mary Johnson	Laptop	10	₦ 450,000	₦ 4,500,000	2025-06-09
S10507	Joy Emmanuel	Laptop	10	₦ 450,000	₦ 4,500,000	2025-03-06
S10691	Blessing Okafor	Laptop	10	₦ 450,000	₦ 4,500,000	2025-08-30
S10766	Anthony Brown	Laptop	10	₦ 450,000	₦ 4,500,000	2025-10-28
S10773	Ibrahim Yusuf	Laptop	10	₦ 450,000	₦ 4,500,000	2025-03-17
S10862	John Williams	Laptop	10	₦ 450,000	₦ 4,500,000	2025-10-02
S10950	Joy Emmanuel	Laptop	10	₦ 450,000	₦ 4,500,000	2025-05-27
S11142	Hauwa Sule	Laptop	10	₦ 450,000	₦ 4,500,000	2025-08-06

Section 4: Five Evidence-Based Business Insights

1. Electronics is the company's primary revenue driver

Electronics generated ₦529.6 million, representing 57.55% of total revenue, making it the dominant category. Laptops alone generated ₦238.05 million.

Business implication: Inventory planning and promotional activities should give strong attention to high-performing electronic products.

2. Revenue performance is highly volatile across months

Revenue peaked at ₦96.356 million in March, representing a 31.17% increase from February. However, revenue subsequently fell by 26.17% in April and by another 29.78% in December.

Business implication: The company needs to investigate the causes of seasonal declines and introduce targeted promotions during weak periods.

3. A relatively small group of customers represents significant value

Joy Emmanuel alone generated ₦64.23 million, while the five highest-spending customers were Joy Emmanuel, Grace Musa, Ruth Adams, Michael James and Mary Johnson.

Business implication: Losing high-value customers could have a significant effect on revenue, making customer retention particularly important.

4. Customers use multiple payment channels

Cash, Mobile Money, Card and Bank Transfer all contributed substantial transaction volumes and revenue, with no single method completely dominating the market.

Business implication: Maintaining multiple payment options can support customer convenience and reduce friction during purchases.

5. Female and male customers contribute almost equally to revenue

Female customers generated ₦467.142 million compared with ₦451.860 million for male customers. The relatively small difference indicates that sales are not heavily dependent on one gender group.

Business implication: Marketing strategies should remain inclusive rather than focusing disproportionately on one gender.

Section 5: Strategic Recommendations

a. Implement CRM Tiering

Establish VIP customer tiers for high-value customers and provide loyalty rewards, personalized offers and priority access to high-demand electronics.

b. Strengthen Electronics Inventory Management

Because Electronics contributes 57.55% of total revenue, management should maintain sufficient stock of high-performing products such as laptops, smartphones and tablets.

c. Develop Seasonal Sales Campaigns

The significant decline in April and especially December indicates the need for targeted promotions, discounts and campaigns during periods of declining sales.

d. Maintain Multiple Payment Options

The company should continue supporting Cash, Mobile Money, Card and Bank Transfer because customers demonstrate demand across all four channels.

e. Use Customer Segmentation for Marketing

Customers should be segmented based on spending and purchasing frequency. High-frequency customers can receive loyalty incentives, while high-spending customers can receive VIP treatment and personalized offers.

f. Monitor Low-Performing Products

Products with relatively low revenue, such as T-Shirt, Cereal and Sugar, should be reviewed to determine whether pricing, demand, inventory levels or promotional strategies need adjustment.

Conclusion

The SQL analysis shows that Keffi Tech Sales generated ₦920.293 million from 1,500 transactions in 2025. Electronics was the strongest category, high-value customers made an important contribution to revenue, and laptops were the leading individual product. However, the business experienced substantial monthly revenue fluctuations, particularly the sharp decline toward the end of the year.

Overall, the findings demonstrate that the company can improve performance by strengthening inventory management for high-performing products, retaining valuable customers, addressing seasonal sales declines, maintaining flexible payment options and using customer data to support targeted marketing decisions.